PH 724 Data Analysis & Scientific Communication Incorporating Use of Generative AI

Part 9: Introduction to Practice Project 3 (the Plight of the Bumblebee, Chp. 9 of [Textbook])

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Scientific background

Bumblebees

Bumblebees are important pollinators in natural ecosystems and agriculture.

Their health may be affected by:

- Habitat conditions.
- Environmental changes.
- Pathogens and parasites.
- Interactions with flowering plants.

The parasite *Crithidia bombi* can infect bumblebees.

Different flowering plants may influence pathogen transmission.

This motivates an experiment examining the relationship between flower type and infection.



Scientific question

Does flower treatment affect the probability that a bumblebee has detectable *Crithidia bombi* infection?

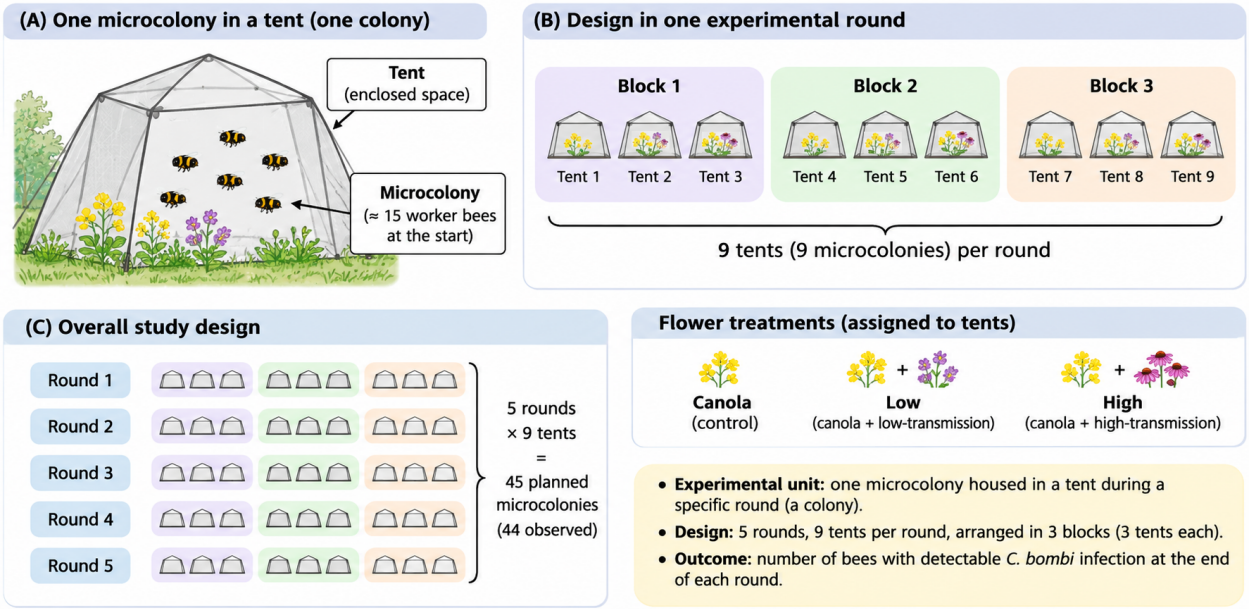
Study design

Microcolonies, tents, and experimental rounds

Bumblebees were housed in microcolonies within tents.

- A microcolony refers to a group of bumblebees living together.
  - Each microcolony initially contained approximately 15 worker bees.
  - Bees within the same microcolony lived together and could interact.
  - Consequently, infection outcomes among bees within the same microcolony might be correlated.
  - The textbook confirms that "All bees were initially infected." (?)
    - Adler et al. (2020) deliberately *inoculating* bees by exposing bees in the infected group to Crithidia bombi, then measured their parasite loads at the end of the experiment. Some surviving bees had zero detectable parasite cells, even though they had received the initial infection treatment.
- A tent was an enclosed experimental space used to house one microcolony.
  - Researchers placed particular flowering plants inside each tent, restricting the bees to foraging on the flowers provided.
  - Nine tents were arranged in three blocks.
  - All bees in the same block come from the same parental lineage.
- The experiment involved three flower treatments. Each tent/microcolony received one flower treatment.
  - Canola: Canola flowers alone (control).
  - Low: Canola combined with flowering plants previously classified as low-transmission.
  - High: Canola combined with flowering plants previously classified as high-transmission.
- Five experimental rounds.
  - Each round lasted two weeks.
  - The same nine tents (at the Amherst Centre for Agriculture) were available per round.

Infection was measured for individual bees at the end of each round (each taking 2 weeks).



Final data

```
##############################
## only using bees in infected treatment
##############################

library(dplyr)
# Load data
cprop <- read.table(
  "https://www.stat.uchicago.edu/~pmcc/projects/ch09/bumblebee/crithidia_infection_data.txt",
  header = TRUE, na.strings = "."
)
treats <- read.table(
  "https://www.stat.uchicago.edu/~pmcc/projects/ch09/bumblebee/flower_strip_treatments.txt",
  header = TRUE, na.strings = "."
)

# Keep only initially inoculated colonies and necessary columns
bees <- cprop %>%
  left_join(
    treats %>% select(tent, round, block, treatment, infect),
    by = "tent"
  ) %>%
  filter(infect == "I") %>% # bees initially inoculated
  transmute(
    colony = factor(tent),
    round = factor(round),
    block = factor(block),
    treatment = factor(treatment),
    infected = infectedbees,
    uninfected = totalbees - infectedbees,
    total = totalbees
  )
```

Aggregated colony-level data

- 44 observed colonies (one planned colony was lost).
- 389 bees assessed at follow-up.
  - There must have been substantial mortality in some tents, but the available counts alone cannot tell us exactly how many bees died, when they died, or whether mortality differed by flower treatment.
- Each row contains the number of infected and uninfected bees in that colony.

Outcomes

The number of infected and uninfected bees within each colony, since one may not distinguish one bee from the other.

Explanatory variables

- Flower treatment
- Round
- Block

- Colony

Assumptions

Biological considerations

- Bees within the same colony share environmental conditions.
- Infection may spread between bees within a colony.
- Colonies may differ in baseline infection risk.
- Infection probabilities may vary between experimental rounds.

Possible sources of variation:

- Bee-level variation: Different infection outcomes among bees within the same colony.
- Colony-level heterogeneity: Colonies may have different underlying infection probabilities.
- Round-level heterogeneity: Infection probabilities may differ between experimental rounds.
- Block-level heterogeneity: Tents in the same design block may share relevant conditions.

Your task: preliminary analysis

- Exploratory data analysis
  - Calculate infection proportions for each colony.
  - Compare infection proportions across treatments.
  - Examine differences between experimental rounds.
  - Check colony sizes and missing observations
- Informative visualizations
  - Colony infection proportions by treatment.
  - Infection proportions by round and treatment.
  - Distribution of colony sizes.
  - Infected and uninfected counts by treatment.
- A proposed plan for formal analysis
  - Choose an appropriate model for infection counts.
  - Account for relevant clustering or overdispersion.
  - Explain the interpretation of treatment effects.

Potental statistical models

Options may include:

- Binomial generalized linear models (used by the textbook)
- Binomial generalized linear mixed-effects models (used by the textbook)
- Quasi-binomial models
- Beta-binomial models

These methods differ in their assumptions about the mean response, dependence, and variation.

The most complicated model is not necessarily the most appropriate model.